

## ***Detection and characterization of Clostridium difficile in retail chicken.***



AIMS: This study was designed to evaluate the prevalence of *Clostridium difficile* contamination of retail chicken. METHODS AND RESULTS: Chicken legs, thighs and wings were purchased using a standardized method from retail outlets across Ontario, Canada. Selective culture was used for qualitative and quantitative detection of *C. difficile*. *Clostridium difficile* was isolated from 26/203 (12.8%) chicken samples; 10/111 (9.0%) thighs, 13/72 (18%) wings and 3/20 (15%) legs ( $P = 0.19$ ). All isolates were ribotype 078, a strain that has been associated with food animals and potentially community-associated disease in humans. All positive samples were positive only on enrichment culture. CONCLUSIONS: *Clostridium difficile* could be found relatively commonly in retail chicken meat, albeit at low levels. SIGNIFICANCE AND IMPACT OF THE STUDY: This is the first study to report *C. difficile* in chicken meat. Contamination of meat with *C. difficile* strains implicated in human infections raises concerns about food as a source of *C. difficile* infection. The relevance of food contamination is completely unclear at this point but food should be investigated as a source of infection.